ERIN CATE LUMSDEN

SOFTWARE ENGINEER

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ABOUT ME

Software engineer with a strong foundation in building reliable, performance-focused systems, with an interest in automotive and motorsport technology. I graduated from RMIT University with a Bachelor of Software Engineering in 2024 and have experience working across application development, real-time data handling, and system design. I am motivated to apply my technical skills in a fast paced environment, contributing to data-driven performance and engineering systems.

EXPERIENCE

Jan 2024 - Current

Firmware & Software Engineer - Brightspace AI

Climate Tech industry startup developing IoT devices that reduce energy usage and improve occupant health. My role spans building custom web tools and on-device processing scripts in C and Python to transmit data across mesh networks, as well as designing microservices and coordinating Docker containers to support uploads of data to the cloud. I have developed HVAC control algorithms that integrate with Building Management Systems (BMS), maintained full-stack web applications including backend services, user and admin frontends, and managed API integrations for external weather and energy monitoring data.

Jan 2022 - Dec 2024

Electronics System Head - RMIT Racing FSAE-A Team

Built custom firmware for vehicle in MoTeC M1 Build, including Traction Control, Throttle Mapping and Auto Shifting. Assisted with vehicle assembly, exhaust repacking, carbon manufacturing and engine tuning via MoTeC M1 Tune. Managed a sub-team of engineering students whilst designing and prototyping components before manufacturing PCBs. Built a motorsport grade harness for competition including custom code for PCBs and intensive testing and problem solving on the go.

Aug 2024 - Nov 2024

Final Year Project - Autonomous Vehicle Datalogger

Built a C# application for an Industry Client. The application connects to an Autonomous Vehicle Datalogger and allows safe scrutineering of the vehicle to deem it's safety condition. Delivered final product to client including test results, automated tests, a release for Linux and Windows, and documentation.

March 2023 - Nov 2024

Software Developer Intern - Nuvotion Pty Ltd

Built and deployed a server to host database and UI for database to manage parts for PCB design and store results from PCB testing. Assisted with electronics manufacturing and assembly.

March 2023 - Jan 2024

Full Stack Software Developer Intern - IDOM Innovations

Working in a team of developers to produce web products in the automotive industry, using a wide range of technology to market and sell vehicles. Worked on internal tools to enhance customer interactions by sending automated SMSs and Emails. Stack included AWS, Ruby, JavaScript & React.

ACHIEVEMENTS

2024 **1st Place - ICE Class FSAE-Australaisa**

- 1st Place Skidpad
- 1st Place Endurance
- 1st Place AutoX
- 1st Place Efficiency
- 1st Place Business
- 1st Place Design

2023 **11th Place - EV Class FSAE-Australasia**

- 11th Place Skidpad
- 7th Place Endurance
- 10th Place AutoX
- 5th Place Efficiency
- 17th Place Business
- 15th Place Design

2022 **2nd Place - ICE Class FSAE-Australaisa**

- 1st Place Endurance
- 1st Place Design
- 1st Place Efficiency

2021 **Westpac Young Technologist Scholarship**

2021 **Telstra Womens Business Circle Education Bursary**

References available upon request.

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GitHub https://github.com/erxncxte

LinkedIn https://www.linkedin.com/in/erin-lumsden/

Portfolio https://erincate.com/